MINISTRY OF EDUCATION AND VOCATIONAL TRAINING

CERTIFICATE OF SECONDARY EDUCATION EXAMINATION

URBAN WEST REGION – ZANZIBAR

FORM FOUR (IV) MOCK EXAMINATION – 2025

BIOLOGY 2A

(For Both School and Private Candidates)

033/2A (ACTUAL PRACTICAL)

TIME: 2:30 HOURS **JUNE 2025**

INSTRUCTIONS:

1. This paper consist of **two (2)** questions
2. Answer **all** questions
3. Each question carries **25** marks
4. Except for diagrams which must be drawn in pencil, all writings must be in **blue** or **black** pen.
5. Calculators, cellular phone and any unauthorized materials are **not allowed** in the examination room
6. Write your **Examination Number** on every page of your answer booklets

0001

1. You are provided with beaker A and B , hot water , cold water and Sample Solution A1 carryout following experiment

Procedures

- Pour about 200cm3 of hot water in the beaker A and Cold water into the beaker B respectively .
- Put three drops of Sample Solution A1 into the beaker A and B containing hot water and Cold water respectively
- Observe Carefully the experiment after five (5) minutes.

QUESTIONS

(a) (i) What happened to the Sample Solution A1 when it was put in hot water and cold water

(ii) Explain why the changes you observed occurred?

(iii) With regard to your obsrevations explain briefly how the beaker A differ from the beaker B after put Sample Solution A1

(b) (i) What do you think was the aim of the experiment .

(ii) Draw a conclusion from the experiment

(iii) Name and define the physiological process investigated in the experiment

(c) (i) State three factors that affect the rate of biological process named in (b)(iii) above

(ii) Outline three significances of the process named above

(iii) Suggest two (2) organs in human being and plants that perform their work by the process named (b)(iii) above

2. You have been provided with Specimen S1 , S2 , S3 , S4 and S5

(a) Using a hand lens , Study the Specimens Carefull and :

(i) Identify each Specimen by its common name

(ii) Classify Specimens S1 and S2 to the class level .

(iii) State three distinctive characteristics that place each of Specimens S1 and S2 into their respective class.

(iv) Give three ways in which Specimen S1 and S2 Differ

(b) (i) Name the organism from which Specimen S3 was taken

(ii) State two adaptations shown by the organism you named in (b)(i) to its mode of locomotion

(iii) What is the function of Specimen S3 to the Organism

(c) (i) What part of Specimen S4 involved in propagation

(ii) Which types of fruit identified by Specimen S5 Give reason

(d) Using a Scalpel/ Knife , make a longitudinal Section of S5 . Draw and label the parts you see.